MIL-A-25949E
AMENDMENT 2
19 JUN 85
SUPERSEDING
AMENDMENT 1
30 January 1980

MILITARY SPECIFICATION

ACCELEROMETER, AIRCRAFT RANGE -5 TO +10g UNITS,
INTEGRALLY LIGHTED

This amendment forms a part of Military Specification MIL-A-25949E, dated 25 April 1978, and is approved for use by all Departments and Agencies of the Department of Defense.

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*Under SPECIFICATIONS, FEDERAL delete "QQ-P-416 Plating Cadmium (Electro-deposited)" and add "QQ-P-601 Boxes, Wood, Cleated Plywood".

*Under SPECIFICATIONS, MILITARY add "MIL-C-83488 Coating, Aluminum, Ion Vapor Deposited".

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*Under STANDARDS, MILITARY add "MIL-STD-129 Marking for Shipment and Storage" and "MIL-STD-794 Parts and Equipment, Procedures for Packaging and Packing of".

*Add:
"2.1.1 Other government documents, drawings and publications.

Air Force Drawing 8346561."

*Add:
"2.1.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issue of this document which is indicated as DOD adopted shall be the issue listed in the current DODISS and supplement thereto, if applicable.

ASTM-D-3951-82 Packaging, Commercial

(Industry associations, specifications and standards, are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies)."

*Add:
2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence."

FSC 6610

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*3.1 Delete and add:
"3.1 First article. This specification makes provisions for first article testing (see 4.4 and 6.3)"

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*3.6.i Delete "0.2g" and add "0.25g".

*3.6.j Delete "±0.2g" and add "±0.25g".

*3.6.k Delete "±0.2g" and add "±0.25g".

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*3.10.2 Delete and add:
"3.10.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum where practical, in accordance with MIL-C-83488 Type I or II as applicable, and of a class that is adequate to achieve the degree of protection required. Other protective coatings, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used."

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*Add:
"3.14 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item."

4.2.a Delete "Preproduction" and add "First article".

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*4.3.2 Delete and add:
"4.3.2 Instrument vibration. Unless otherwise specified, the accelerometer shall be vibrated before a test reading is taken. This vibration shall be accomplished utilizing an electric motor and an attached counterweight as defined by Air Force Drawing 8346561".

*Add:
"4.3.5 Allowable centrifuge vibration. The centrifuge utilized for accomplishing the tests herein shall not have an inherent vibration of more than $0.04q^2$ between 2 and 200 cps where "g" is defined as $g=0.0511\ f^2D$ with "D" being displacement in inches (double amplitude) and "f" is the frequency in cycles per second. Inherent vibration shall be defined as the vibration sensed by the accelerometer which is induced by the centrifuge when testing in accordance with this document. Installed on the centrifuge shall be a centrifuge rotor per Air Force Drawing 8346561 for mounting the accelerometer for test."

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*4.4 Delete "Preproduction" and add "First article".

*4.4.1 In sixth line delete "qualification" and add "first article".

*4.4.3 Delete and add:
"4.4.3 First article tests. The first article tests shall consist of all the tests specified under 4.6, and the first article phase of 4.7."

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*4.5.1.f Delete "4.6.5.1" and add "4.6.5.3".

4.5.1.g Delete and add:
"4.5.1.g Leakage (ABU-4B/A only)----------see 4.6.19"

4.5.1.h Delete and add:
"4.5.1.h Fogging (ABU-4B/A only)----------see 4.6.20"

*4.5.2.1.c Delete and add:
"4.5.2.1.c Force on knob----------see 4.6.5.2"

*Add:
"4.5.2.1.i Locking mechanism operation----------see 4.6.5.1"

4.5.2.2.1 Delete and add:
"4.5.2.2.1 Internal examination----------see 4.6.21"

*4.6.2.c Change the last two sentences to read: "The static scale error shall not exceed ±0.25g units in any of the three positions tested. Friction shall not exceed ±0.25g units at the a. (1g) and c. (-1g) test positions and ±0.3g units at the b. (0g) test position with the memory pointers engaged."

*4.6.2.1 Delete and add:
"4.6.2.1 Dynamic scale error and friction. The accelerometer shall be mounted on a suitable centrifuge and centrifuge rotor per Air Force Drawing 8346561 and tested for dynamic scale error and friction for accelerations of -5, -4, -3, -2, +2, +3, +4, +5, +6, +7, +8, +9 and +10 at 1g increments for a total of 13 test points. For the first test point and without instrument vibration, the centrifuge shall be brought up to, but not exceeding, the rotation rate equivalent to the "g" level being tested. The centrifuge rate shall be reduced to zero and the accelerometer reading taken from the memory pointer. After re-zeroing the memory pointer, the procedure shall be repeated except that instrument vibration shall be applied per paragraph 4.3.2. The centrifuge rate shall be reduced to zero and the accelerometer reading taken from the memory pointer. The memory pointer shall then be zeroed. The reading with and without vibration shall not differ from the centrifuge input "g" reading by more than ±0.25g. This procedure shall be repeated for the remaining 12 test points. The centrifuge rate need not be returned to zero and the memory pointer reset, if capability exists to read the instruments without reducing the centrifuge's rotational rate."

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*4.6.3.1.1 In last sentence delete "Preproduction" and add "First article".

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*Add:
"4.6.5.3 Locking mechanism operation (individual tests).

a. Manually shake the accelerometer to display the auxiliary pointer to at least +4g and -4g.

b. Lock the accelerometer and shake the accelerometer to the approximate same level with auxiliary pointers still in position as in "a" above. Verify that the mechanism is locked, which will be indicated by the main pointer not moving during the shaking motion.

4.6.6 In fourth line delete "4.6.3" and add "4.6.2.1".

4.6.7 In fourth line delete "4.6.3" and add "4.6.2.1".

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4.6.10 In eleventh line delete "4.6.3" and add "4.6.2.1".

4.6.11 Delete and add:
"4.6.11 Shock. The accelerometer shall be mounted in the normal operating position with the mechanism locked and subject to a shock test in accordance with MIL-STD-810, Method 516.2, Procedure I, Figure 516.2-2. The shock pulse shall be half sine wave form with a peak value of 15 g's for a time duration of 11 milliseconds. After the completion of the cycle, the units shall be examined for damage and then subjected to the tests specified in 4.6.2.1."

4.6.11.1 Delete

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4.6.13 In third line delete "th" and add "the". In fourth line delete "4.6.3" and add "4.6.2.1".

4.6.19 In title delete "AFB" and add "ABU".

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*4.7.1 In title delete "Qualification" and add "First article". In tenth line delete "4.6.3" and add "4.6.2.1".

4.7.3 In fifth line delete "4.6.3" and add "4.6.2.1".

*4.8 Delete and add:
"4.8 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with MIL-P-116 to insure compliance with Section 5 of this specification".

*5.1 Delete "or C" and add "C or industrial".

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*Add:
"5.1.3 Industrial preservation. The industrial preservation of accelerometers shall be in accordance with ASTM-D-3951-82."

*5.2 Delete "or C" and add "C or industrial".

*5.2.1 In line two delete "PPP-B-636, weather resistant class" and add "PPP-B-601, Styles A or B, class overseas".

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*5.2.2 At the end of the first sentence add ", class weather resistant." Delete last sentence.

*5.2.3 In last sentence place a period after "regulations" and delete rest of sentence.

*Add:
"5.2.4 Industrial packaging. The packaged accelerometers shall be packed in accordance with ASTM-D-3951-82."

*5.4 Delete and add:
"5.4 Marking. In addition to any special marking required by the contract or order, each unit package and shipping container shall be marked in accordance with MIL-STD-129."

*Add:
"5.4.1 Precautionary marking. The following precautionary marking shall appear conspicuously on two opposite sides of each interior package and shipping container, wherever applicable:

FRAGILE-DELICATE INSTRUMENTS
HANDLE WITH CARE"

*6.3 In title delete "Preproduction" and add "First article".

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The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Air Force-99
Navy-AS

Preparing activity:
Air Force-71

Project number:
6610-0091

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